Indian Institute of Technology Gandhinagar
Department of Mathematics

MA 104: Ordinary differential equations
(Course plan, Semester II, 2025 - 26)

Instructor: Sudeshna Roy

Email: sudeshna.roy@iitgn.ac.in

Office: AB 06/347

TA: TBA

Course Contents

- First order ODE: geometric meaning, Direction fields, Linear equations
- Separable and exact equations, Integrating factors, Bernoulli equations
- Orthogonal trajectories
- Lipschitz condition, Picard's theorem, Examples on non-uniqueness
- Second order linear differential equations
- Linear independence of solutions and the Wronskians
- Dimensionality of space of solutions, Abel-Liouville formula
- Linear ODEs with constant coefficients, the characteristic equations
- Cauchy-Euler equations
- Method of undetermined coefficients
- Method of variation of parameters
- Higher order linear ODE: general principles
- Power series solutions of ODE's
- Legendre's equation and Legendre's polynomials
- Regular and irregular singular points, method of Frobenius
- Bessel's equation and Bessel's functions
- Laplace transform: inverse transform, Linearity
- Transform of derivatives and integrals, Differential equations
- Unit step functions, Dirac's delta function, Shifting theorems
- Differentiation and integration of transforms
- Convolution theorem, Integral equations

Textbook:

- E. Kreyszig, Advanced Engineering Mathematics (10th Edition), John Wiley, 2011.

Reference books:

- W. E. Boyce and R. DiPrima, Elementary Differential Equations (8th Edition), John Wiley, 2005.
- G.F. Simmons, Differential Equations with Applications and Historical Notes, Tata McGraw-Hill Publishing Company Limited, 1974.

Academic integrity

The students must adhere to the norms of honour code in the quiz/exam. Violation of academic integrity like, dishonesty in examination or any other academic performances will be reported to Academic office for appropriate disciplinary action accordance with norms of the institute.

Policy for continuous evaluation (tentative)

Attendance	Assignments	Quizzes	Final Exam
10%	10%	40%	40%

Policy for Grading

Relative grading policy will be followed.

Detailed course plan

Topics	Lectures
First order ODE: geometric meaning, Direction fields, Separable and exact equations	1
Integrating factors, Bernoulli equations, Orthogonal trajectories	1
Lipschitz condition, Picard's theorem, Examples on non-uniqueness	1
2nd order linear differential equations: equations with constant coefficients, Euler Cauchy equations	1
The Wronskian, Non-homogeneous ODEs, Method of undetermined coefficients	2
Abel-Liouville formula, Method of variation of parameters, Higher order ODEs	1
Series solutions: Legendre's equation and the Legendre polynomials	2
Regular and irregular singular points, Frobenius' method, the Bessel's functions	2
Laplace transform, Generalities, Shifting theorems, Dirac's Delta function	3